

Koketso Sethosa

Exercise 4-SQL UNION

1) SELECT account_id, account_holder, ~~city~~ 'sandton' AS city
FROM branch_sandton_accounts
UNION
SELECT account_id, account_holder, ~~city~~ 'rosebank'
FROM branch_rosebank_accounts ;

Output Table :

account_id	account_holder	city
A001	Nomvula Dlamini	sandton
A002	David Mokoena	sandton
A003	Lerato Sithole	sandton
A004	Peter Nkosi	sandton
A005	Zanele Khumalo	rosebank
A006	Thabo Mathera	rosebank

```

2) SELECT product_code,
           product_name,
           'savings' AS product product_type
FROM savings_products
UNION
SELECT product_code,
           product_name,
           'current' AS
FROM current_products ;

```

Output Table

product_code	product_name	product_type
SV 01	Basic Savings	savings
SV 02	Premium Saving	savings
SV 03	Youth Saving	savings
SV 04	Business Savings	savings
CR 01	Standard Current	current
CR 02	Gold Current	current
CR 03	Business Current	current

```

3) SELECT staff_id,
        staff_name,
        'Retail AS email Banking' AS email
FROM retail-banking-staff
UNION
SELECT staff_id,
        staff_name,
        'Corporate Banking'
FROM corporate-banking-staff ;

```

Output Table

staff_id	staff_name	email
S001	Mpho Randebe	Retail Banking
S002	Brain Tshabalala	Retail Banking
S003	Aisha Patel	Retail Banking
S004	Kabelo Moabelo	Retail Banking
S005	Nandi Dube	Corporate Banking
S006	Sipho Khumalo	Corporate Banking

```

4) SELECT  city-code,
           city-name,
           'mobile' AS region
FROM mobile-branch-cities
UNION
SELECT  city-code
       city-name
       'digital'
FROM digital-branch-cities;

```

Output Table

city-code	city-name	region
C01	Johannesburg	mobile
C02	Pretoria	mobile
C03	Cape Town	mobile
C04	Durban	mobile
C05	Polokwane	digital
C06	Port Elizabeth	digital

```

3) SELECT customer_id,
        customer_name,
        segment 'push notification' AS segment
FROM push-notification-targets
UNION
SELECT customer_id,
        customer_name,
        'inapp banner'
FROM inapp-banner-targets;

```

Output Table

customer_id	customer_name	segment
C1001	Nomsa Zwane	push notification
C1002	Andile Buthelezi	push notification
C1003	Fatima Mahomed	push notification
C1004	Ryno van Zyl	push notification
C1005	Thandeka Cele	inapp banner
C1006	Samuel Nkosi	inapp banner

```

6) SELECT transaction_id,
      account_id,
      amount,
      transaction_date
FROM atm01_transactions
UNION ALL
SELECT transaction_id,
      account_id,
      amount,
      transaction_date
FROM atm02_transactions;

```

Output Table

transaction_id	account_id	amount	transaction_date
T1001	A001	500,00	2026-05-24
T1002	A002	120,00	2024-0 ⁵ 4 -26
T1003	A001	300,00	2026-05-2 4 ⁶
T1004	A003	750,00	2026-05-28
T1005	A004	900,00	2026-05-28
T1006	A002	450,00	2026-05-31
T1007	A005	150,00	2026-05-31
T1008	A001	300,00	2026-05-26


```

7) SELECT application-id, app-id AS application-id,
        customer-id,
        loan-type,
        amount-requested
FROM gauteng-loan-applications
UNION ALL
SELECT app-id,
        customer-id,
        loan-type,
        amount-requested
FROM western-cape-loan-applications;

```

Output Table

application-id	customer-id	loan-type	amount-requested
LA001	C1001	Home Loan	2500
LA002	C1002	Vehicle Loan	1500
LA003	C1003	Personal	1000
LA004	C1004	Home Loan	3000
LA003	C1003	Personal	1000
LA005	C1005	Vehicle Loan	2500
LA006	C1006	Business	1800
LA007	C1007	Home Loan	2200

```

3) SELECT complaint-id,
      customer-id,
      category,
      logged-date
FROM   email_complaints
UNION ALL
SELECT complaints-id,
      customer-id,
      category,
      logged-date
FROM   app_complaints ;

```

Output Table

complaint-id	customer-id	category	logged-date
EC 001	C2001	email	2026-04-16
EC 002	C2002	email	2026-04-18
EC 003	C2003	email	2026-04-22
EC 004	C2004	email	2026-04-26 18
AC 001	C2005	app	2026-04-08
AC 002	C2001	app	2026-04-11
AC 003	C2006	app	2026-04-13
AC 004	C2007	app	2026-04-18


```

9) SELECT payment-id,
        account-id,
        amount,
        payment-date
FROM april-payments
UNION ALL
SELECT payment-id,
        account-id,
        amount,
        payment-date
FROM may-payments ;

```

Output Table

payment-id	account-id	amount	payment-date
PAY001	A001	12500,00	2026-04-02
PAY002	A002	4800,00	2026-04-06
PAY003	A003	9200,00	2026-04-11
PAY004	A004	3300,00	2026-04-18
PAY005	A001	12500,00	2026-05-02
PAY006	A005	7600,00	2026-05-08
PAY007	A002	5100,00	2026-05-13
PAY008	A006	2800,00	2026-05-15

10) ~~entry~~ SELECT entry-id,
 account-id,
 entry-type,
 amount,
 entry-date
 FROM debit-entries
 UNION ALL
 SELECT entry-id,
 account-id,
 entry-type,
 amount,
 entry-date
 FROM credit-entries ;

Output Table :

entry-id	account-id	entry	amount	entry-date
DR001	A001	Debit	1000	2026-02-22
DR002	A002	Debit	1500	2026-02-22
DR003	A003	Debit	1800	2026-02-22
DR004	A004	Debit	2000	2026-02-22
CR001	A001	Credit	1000	2026-02-22
CR002	A005	Credit	1800	2026-02-22
CR003	A002	Credit	1500	2026-02-22
CR004	A006	Credit	2000	2026-02-22

Bonus Questions

- 1) Use ~~to~~ UNION because the analyst need to return only unique list of customers from ~~either tables~~ both tables
- 2) UNION ALL because the team wants to return every ~~records~~ transaction that passed even if its a there is duplicates. in coming from 2 systems.
- 3) The first table ~~does not~~ and second table do not have the same number of columns.

Second table has ~~city~~ as a ~~etc~~ column and first table does not have it.

Fix : add city as a column to ~~table~~ first table

Example :

```
SELECT account-id, account-holder, city  
FROM branch-sandton-accounts  
UNION  
SELECT account-id, account-holder, city  
FROM branch-rosebank-accounts
```